

CA – 2nd

CAP538	ALGORITHM DESIGN AND ANALYSIS
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QUESTION : -

Fractional Knapsack

CODE

```
#include <iostream>
#include <algorithm>
using namespace std;

double FractionalKnapsack(int value[], int weight[], int n, int w){
   double ratio[n];

   for(int i = 0; i<n; i++){
      ratio[i] = (double)value[i] / weight[i];
   }
   for(int i = 0; i<n; i++){
      for(int j = i+1; j<n; j++){
         if(ratio[i] < ratio[j]){
            swap(ratio[i], ratio[j]);
            swap(value[i], value[j]);
            swap(weight[i], weight[j]);

         }
      }
   }
   double  totalvalue = 0.0;
```

```
    for(int i = 0; i<n; i++){
        if(weight[i] <= w){
            totalvalue += value[i];
            w -= weight[i];
        }else{
            totalvalue += ratio[i] * w;
            break;
        }
    }
    return totalvalue;
}

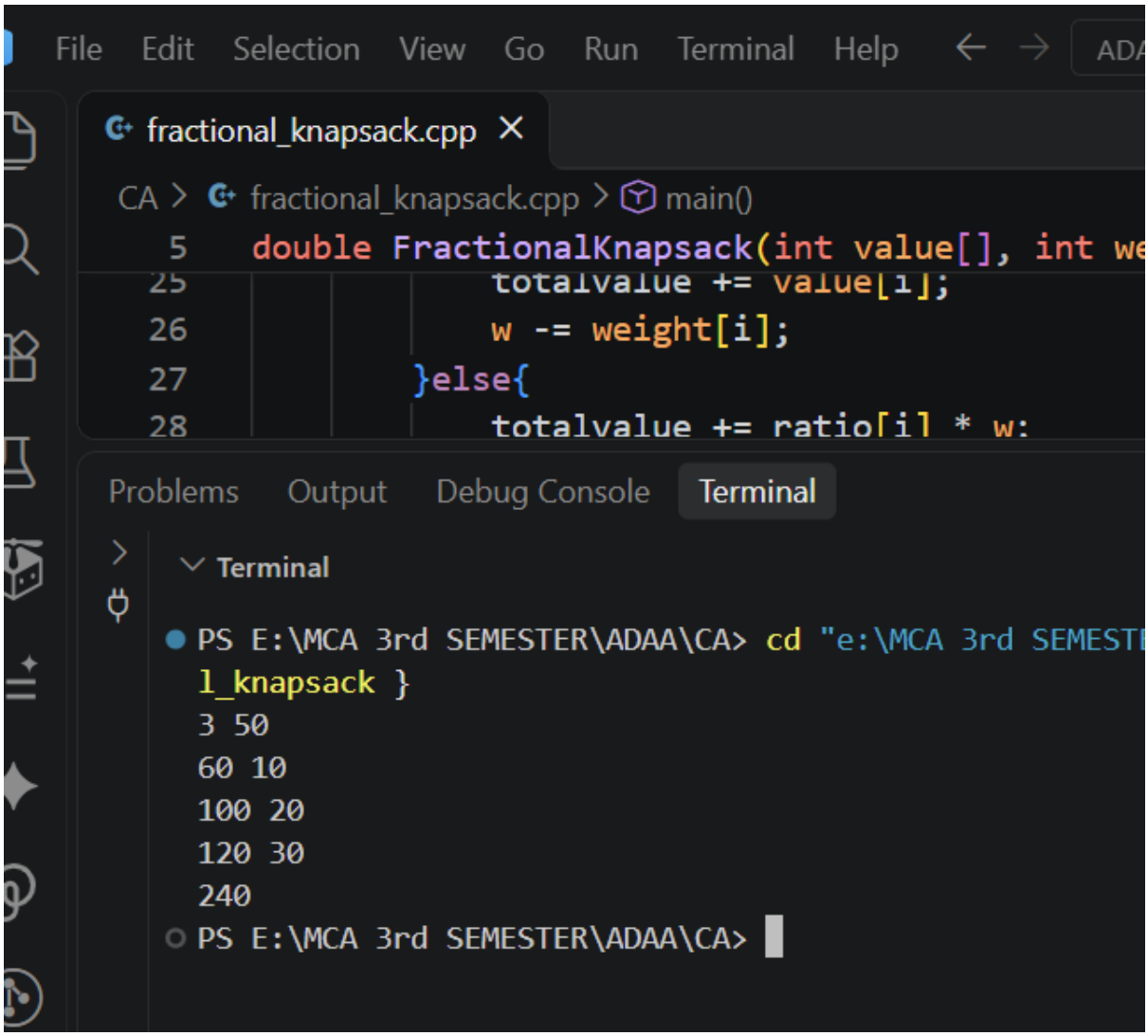
int main(){
    int n;
    int w;
    cin>>n;
    cin>>w;

    int value[n];
    int weight[n];

    for(int i = 0; i<n; i++){
        cin>> value[i] >> weight[i];
    }

    cout<< FractionalKnapsack(value, weight, n, w);
    return 0;
}
```

OUTPUT


File Edit Selection View Go Run Terminal Help
fractional_knapsack.cpp
CA > fractional_knapsack.cpp > main()
5 double FractionalKnapsack(int value[], int we
25 totalvalue += value[i];
26 w -= weight[i];
27 }else{
28 totalvalue += ratio[i] * w;
Problems Output Debug Console Terminal
Terminal
PS E:\MCA 3rd SEMESTER\ADAA\CA> cd "e:\MCA 3rd SEMEST
l_knapsack }
3 50
60 10
100 20
120 30
240
PS E:\MCA 3rd SEMESTER\ADAA\CA>